Production RAG — Complete Results

An evaluation-first retrieval baseline over 202 official OpenAI and Anthropic documentation pages. Every hypothesis tested, including the two that failed.

WHAT THIS PROJECT HAS ACTUALLY ESTABLISHED

Two architectural hypotheses were diagnosed, then tested with controlled interventions, and **both were falsified**. Retrieval did not improve because the diagnoses were wrong — and the value here is that the measurements say so plainly.

The closed-book control **never ran** (no generation credential, provider host egress-blocked), so the project's primary question — does retrieval beat what the model already knows? — remains **unanswered**.

0.000 → 0.475
Lexical baseline, before and after diagnosing why it retrieved nothing

0 rescued
Questions recovered by bounding chunk size (EXP-005), despite a 16,096 → 1,999 char cut

Δ 0.000
Macro recall change from contextual enrichment, isolated (EXP-006 A→B)

The investigation, in order

Initial belief	the shipped lexical baseline works → it returned **nothing at all**
Fix	BM25 over the same index → 0.000 → 0.475
Diagnosis 1	oversized chunks hide evidence → **EXP-005: 0 rescued. Falsified.**
Diagnosis 2	missing structural context → **EXP-006: Δ0.000. Falsified.**
Surviving	BM25 cannot bridge the question/document vocabulary gap → untested

Setup

Corpus	202 documents / 14,209 chunks — Anthropic 139 docs (12,028 chunks), OpenAI 63 docs (2,181 chunks)
Golden set	22 cases: 20 retrieval-scored + 2 abstain controls; 22 evidence spans
Evidence anchoring	`(version_id, section_path, char_start, char_end)` — never chunk IDs, so re-chunking cannot invalidate the benchmark
Retrieval	BM25, `k1=1.2`, `b=0.75`, `simple` config, `top_k=10` throughout
Closed-book control	**blocked — OpenAIError: Missing credentials. Please pass an `api_key`, `workload_identity`, `admin_ap…**

V1 — retrieval methods (EXP-000 … EXP-003)

Experiment	macro recall	fully recalled	spans	doc recall
EXP-000 lexical — *as shipped*	**0.000**	**0/20**	**0/22**	**0.000**
EXP-000 lexical — BM25 (fixed)	0.475	9/20	10/22	0.818

Experiment	macro recall	fully recalled	spans	doc recall
EXP-001 dense (LSA substitute)	0.300	5/20	7/22	0.773
EXP-002 hybrid interleave	0.450	8/20	10/22	0.818
EXP-003 RRF `rrf_k=60`	0.500	9/20	11/22	0.818
EXP-003 RRF pool 100, `rrf_k=60` (tuned on eval set)	0.600	11/20	13/22	0.818

The shipped lexical baseline retrieved nothing at all.

`websearch_to_tsquery` ANDs every token, so a 16-word question required all 16 tokens inside one chunk. Zero hits on all 20 questions. Replaced with BM25 over the same TSVECTOR/GIN index.

Lexical beat the available dense retriever, and hybrid interleave was worse than lexical alone.

BM25 0.475 vs LSA 0.300; switching to dense regressed 5 of 20 questions. Interleave scored 0.450 — below lexical. The dense retriever is an offline TF-IDF+SVD substitute, not a pretrained model; this says nothing about dense retrieval generally.

RRF's gain over lexical is partial credit, not a win.

Per question it is 1 rescued / 1 regressed — a wash. It regressed OA-004, whose evidence sits at lexical rank 5 and dense rank 61; fusion averaged it to rank 13, outside k.

Every configuration hit the same ceiling.

Document-level recall is 0.818 for lexical, interleave and RRF alike, against span recall of 0.455–0.500. Of 12 missed spans, 8 had the correct document in the top 10. That gap set up the next two experiments.

EXP-005 — is chunk granularity the bottleneck?

Hypothesis: if oversized chunks hide evidence, bounding chunk size should move span recall toward the 0.818 document ceiling.

Chunker	chunks	mean	p90	max	>2000 chars
v1 control	14,209	1,097	3,466	**16,096**	3,069
v2 bounded	20,526	757	1,193	**1,999**	**0**

Configuration	macro recall	fully recalled	doc recall	rescued / regressed
EXP-000 control (v1 chunker)	0.475	9/20	0.825	—
EXP-005A bounded (v2)	0.500	9/20	0.825	0 / 0
EXP-005B technical (v3)	0.650	12/20	0.900	3 / 0

FALSIFIED

The intervention was real — the corpus maximum fell from 16,096 to 1,999 characters and all 3,069 over-2,000 chunks disappeared — and it rescued **zero** questions. Across 22 spans it improved 8 ranks and worsened 9, and **not one** previously unreachable span became reachable.

The case that motivated the hypothesis, AN-003, went from 1.65% to 4.79% of its chunk — a 2.9× improvement on exactly the diagnosed quantity — and remained unretrieved.

Smaller chunks also cost real ranks.
AN-002 fell from rank 27 to 172 when splitting cut query-term coverage in its chunk from 12/13 to 7/13. The control's oversized chunks were accidentally *helping* a bag-of-words retriever by aggregating co-occurring terms.

V3's +3 was confounded.
It changed boundaries *and* prepended structural context to the indexed text, and its document recall rose to 0.900 when a pure chunking change should have left it flat. That confound is what EXP-006 was built to resolve.

EXP-006 — is it the structural context?

A 2×2. B and D are row-for-row copies of A and C — **0** boundary differences, **0** body differences — so an A→B difference cannot be a chunking difference. The canonical chunk body is never mutated; the header lives in a separate `search_text` column, because a citation must quote real source text.

Configuration	macro recall	fully recalled	doc recall	MRR	absent@10	absent@300
A — control chunking, plain	0.475	9/20	0.825	0.280	12	1
B — control chunking, enriched	0.475	9/20	0.875	0.282	12	1
C — bounded chunking, plain	0.500	9/20	0.825	0.287	11	1
D — bounded chunking, enriched	0.550	10/20	0.850	0.279	10	0

Comparison	Δ macro	rescued	regressed	net
A->B enrichment on control chunking	+0.000	AN-004	AN-005	+0
A->C chunk size without enrichment	+0.025	—	—	+0
C->D enrichment on bounded chunking	+0.050	AN-004, AN-007	AN-005	+1
B->D chunk size with enrichment	+0.075	AN-007	—	+1

FALSIFIED

A→B is the comparison that isolates enrichment, and it moved macro recall by **exactly zero**. Its single rescue (AN-004, rank 12→7) crossed the cutoff while its BM25 score *fell* — competitors merely lost more. Its regression (AN-005, 4→18) is large.

Mechanism: enrichment inflates document frequency. Writing `Provider: anthropic` into all 12,028 Anthropic chunks took that term from df 3,289 to 12,028 (+266%); `OpenAI` 366 → 2,185 (+497%). BM25 weights by `ln(1+(N−df+0.5)/(df+0.5))`, so a constant field repeated everywhere attacks the very signal it adds. Across all 22 spans enrichment supplied a query term to only **3**, and in **zero** cases a discriminative one.

Exploratory: which header fields matter (dev-set selected, not held out)

Variant	Header fields	macro recall	fully recalled
A	none (baseline)	0.475	9/20
E1_section_only	control chunking, section path only	0.525	10/20
E2_document_section	control chunking, document title + section path	0.475	9/20
B	provider + document + section	0.475	9/20

The ordering matches the mechanism exactly: the more constant the field, the more df inflation and the worse the result. Only the section path — which actually varies chunk to chunk — carries information, and it improves the baseline by **one case**.

AN-003 — the canonical failure, across every experiment

Query: "How many requests can a single Message Batches create request contain at most?" **Evidence:** *"There is a limit of 100,000 messages in a single request."*

Configuration	rank	reachable at depth 300?	document rank
A — control chunking, plain	—	no	6
B — control chunking, enriched	—	no	9
C — bounded chunking, plain	—	no	4
D — bounded chunking, enriched	74	yes	12

Enrichment built a lexical bridge for the first time — rank 74 under D, having been absent at depth 300 everywhere else — because the header supplied `Batches`. Still nowhere near k=10. The terms that would actually identify this evidence appear nowhere in the chunk that answers it: `contain` (df 111), `most` (533), `many` (564), and `requests`, which never matches the body's singular `request` because the deliberately unstemmed `simple` configuration cannot bridge it.

This is a **vocabulary mismatch**, and it is why both architectural hypotheses failed.

Limitations — read before quoting any number

1. **The closed-book control never ran.** No generation credential and the provider host is egress-blocked. Retrieval is uncalibrated against what the model already knows — the project's primary question is still open.
2. **Dense retrieval has not been tested.** The dense numbers use an offline TF-IDF+SVD substitute; both the sentence-transformer host and the OpenAI embedding endpoint are egress-blocked. All dense, hybrid and RRF figures are lower bounds, and nothing here supports "BM25 beats dense retrieval".
3. **n = 20 is development scale.** One case is 5 points of macro recall. Every EXP-005/006 movement is 1–2 cases. Nothing is statistically significant; the paired per-case tables are the trustworthy part.
4. **Some numbers are eval-set tuned** — the RRF pool-100 configuration and the EXP-006 field ablation. Both are labelled where they appear. BM25 `k1` and `b` were never swept.
5. **V3's largest rescue is still unexplained.** AN-010 (rank 49→1) is reproduced by no EXP-006 configuration, so it likely came from V3's table row-group splitting, which was never isolated.
6. **Corpus skew:** 139 Anthropic documents to 63 OpenAI, because OpenAI's own documentation hosts were unreachable and only its public repositories could be used.
7. **The golden set is single-author.** Anchors were mechanically verified against source, but no second annotator reviewed them.

What the data justifies next

A real pretrained embedding model. Both lexical hypotheses have been tested and rejected by controlled intervention. The surviving diagnosis — vocabulary mismatch — is exactly what semantic retrieval addresses, and AN-003 is its canonical test case: an answer whose discriminative query terms appear nowhere in the text that answers it. Blocked only on network egress.

Secondary and cheap: isolate V3's table row-group mechanism, the last unexplained rescue; and add stemming as a *third ranked list* rather than a replacement, so identifier precision survives while plural/singular matching is recovered.

Still not justified: a reranker — AN-003 is absent from depth 300 in three of four configurations; reranking reorders candidates, it cannot retrieve missing evidence. Nor freezing enrichment, nor a larger k, nor a confidence threshold.

Baseline decision: control chunking, unenriched, BM25. Neither new chunker nor enrichment earned promotion.

Generated from experiments/*.json by scripts/build_consolidated_pdf.py. Git commit 3843627979fd. EXP-006 config hash f603aaac10c0e5fe. Corpus snapshot snap_689e3363…. Per-experiment reports: EXP-005-rechunking.pdf, EXP-006-enrichment-ablation.pdf. Raw provider documentation is not redistributed; published results carry evidence anchors, ranks and scores with retrieved text replaced by a content hash.